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$$w = e^{-x^2-y^2}$$

$$x = t$$

$$y = \sqrt{t}$$

$$\frac{\partial w}{\partial x} = -2xe^{-x^2-y^2}$$

$$\frac{\partial w}{\partial y} = -2ye^{-x^2-y^2}$$

$$\frac{dx}{dt} = 1$$

$$\frac{dy}{dt} = \frac{1}{2\sqrt{t}}$$

$$\frac{\partial w}{\partial t} = -2xe^{-x^2-y^2} \cdot (1) + -2ye^{-x^2-y^2} \cdot \left(\frac{1}{2\sqrt{t}}\right)$$

$$= -te^{-t^2-t} - e^{-t^2-t}$$

$$= (-2t-1)e^{-t^2-t}$$

References